Secondary Commercial Real Estate Market Analysis in Krakow

Magda Zielant

Overview

The analysis focuses on the commercial real estate market (retail and service properties).

Data Scope:

- **Location:** *City of Krakow* (diverse districts, ranging from the strict city center to intermediate and peripheral zones).
- **Market Type:** Secondary market (purchase and sale transactions of existing properties).
- **Timeframe:** Transactions concluded between January 2017 and April 2018.
- **Subject of Analysis:** Commercial premises with a usable area between 60 and 85 sqm, located primarily on the ground floors, basements, and upper floors of multi-family buildings and tenement houses.

An analysis of transactions from 2017–2018 reveals three main zones with varying potential:

- **Downtown and Tourist Zone** (**Old Town, Kazimierz**)
  - *Streets:* Długa, Stradomska, Starowiślna, Dajwór, Tokarzewskiego.
  - *Characteristics:* The area with the highest prestige and property values. Foot traffic is driven primarily by tourists and premium services (gastronomy, short-term rentals).

- **Intensive Urbanization and Academic Zone** (**Krowodrza, Grzegórzki**)
  - *Streets:* Kijowska, Lea, Czarnowiejska, Mogilska, Grzegórzecka.
  - *Characteristics:* Densely populated districts catering to students and corporate professionals. Commercial spaces primarily serve permanent residents. There is a visible increase in market attractiveness driven by new office developments.

- **Intermediate and Residential Zone** (**Prądnik, Podgórze, Bieżanów**)
  - *Streets:* Dobrego Pasterza, Śliczna, Wielicka, Cechowa, Bieżanowska.
  - *Characteristics:* Lower price points, with demand strictly dependent on local neighborhood demographics. This zone carries a higher risk of vacancy compared to the city center.

An economic analysis of Krakow highlights three decisive factors influencing the commercial real estate market:

- **Strong Tourist Influx:** A record number of visitors (approx. 12.9 million in 2017) directly increased prices and demand for food and beverage premises in the city center, notably on streets like Stradomska and Dajwór.

- **Labor Market and Purchasing Power:** A drop in unemployment to approx. 3% (creating an employee's market) combined with wage growth significantly boosted residents' consumption. This stabilized retail turnover and the commercial rental market.

- **Stable Interest Rates:** The historically low reference rate of the National Bank of Poland (1.5%) favored affordable credit, positioning real estate as a highly attractive investment alternative to traditional bank deposits.

Data Analysis & Results

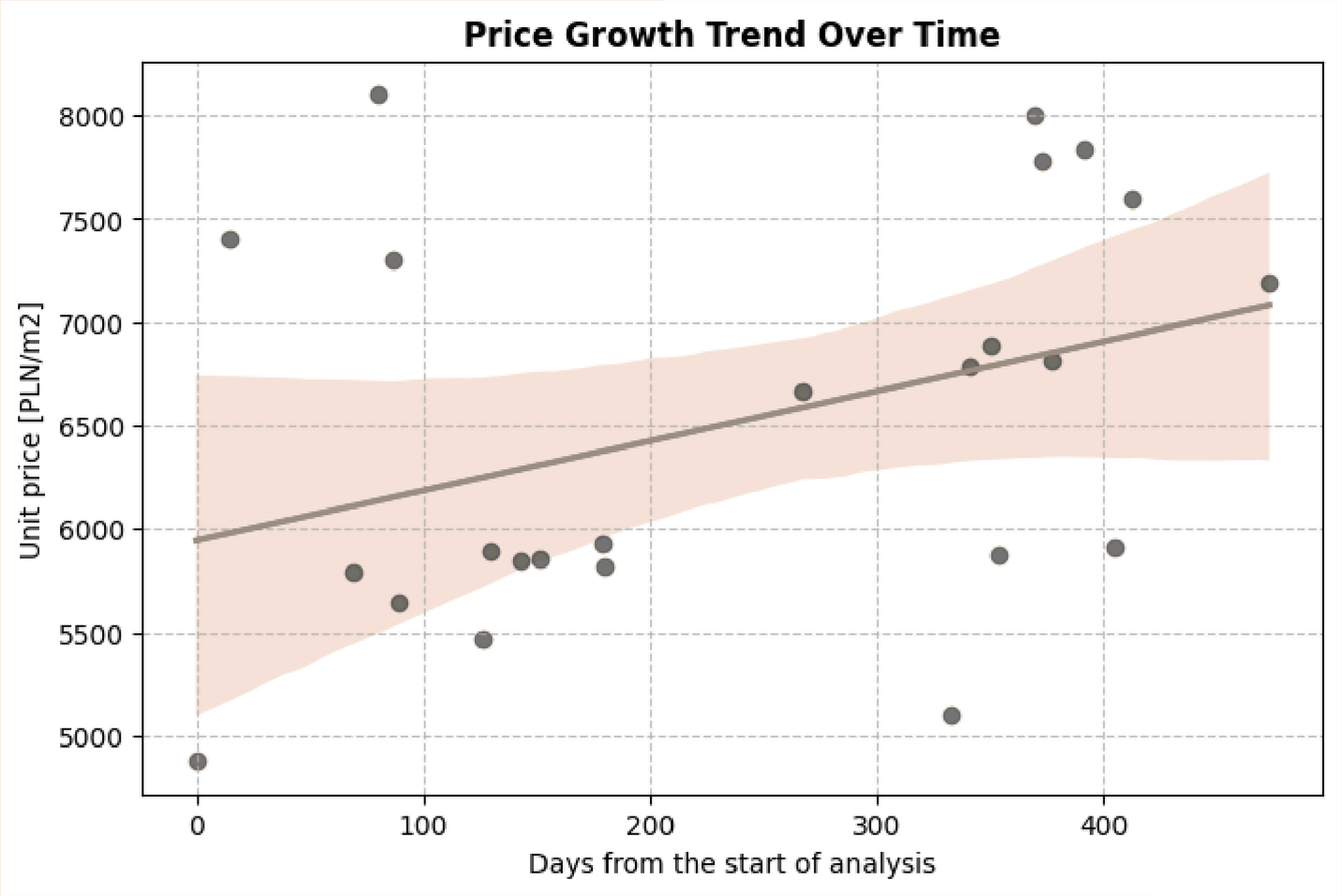
1. Price Trend Over Time
2. Regression: Property Price vs. Area
3. Market Segmentation
4. Valuation Model
5. Feature Interdependence

Tech Stack & Libraries

- pandas
- numpy
- matplotlib.pyplot
- seaborn
- statsmodels.api
- sklearn.linear_model (LinearRegression)
- sklearn.cluster (KMeans)
- sklearn.preprocessing (StandardScaler)

Descriptive Statistics for Price per m^2

count	24.000000
mean	6514.611667
std	964.881494
min	4876.150000
25%	5835.145000
50%	6295.555000
75%	7326.542500
max	8100.490000



Determining the Price Trend Over Time

Investigated whether the passage of time during the analyzed period (1.5 years) impacted property values.

- **Method:** Linear regression of the Unit Price against the Transaction Date.
- **Result:** The trend slope coefficient is +2.40 PLN/day.
- **Conclusion:** The market was in a growth phase. The estimated nominal annual increase in transaction prices was approximately 878 PLN per square meter.

Implementation

```
model_trend = LinearRegression()
model_trend.fit(df[['Date_num']], df['Unit Price [PLN/m2]'])

daily_trend = model_trend.coef_[0]
print(f"\nTREND: Daily price increase: {daily_trend:.2f} PLN/m2")
print(f"Estimated annual increase: {daily_trend * 365:.2f} PLN/m2")
```

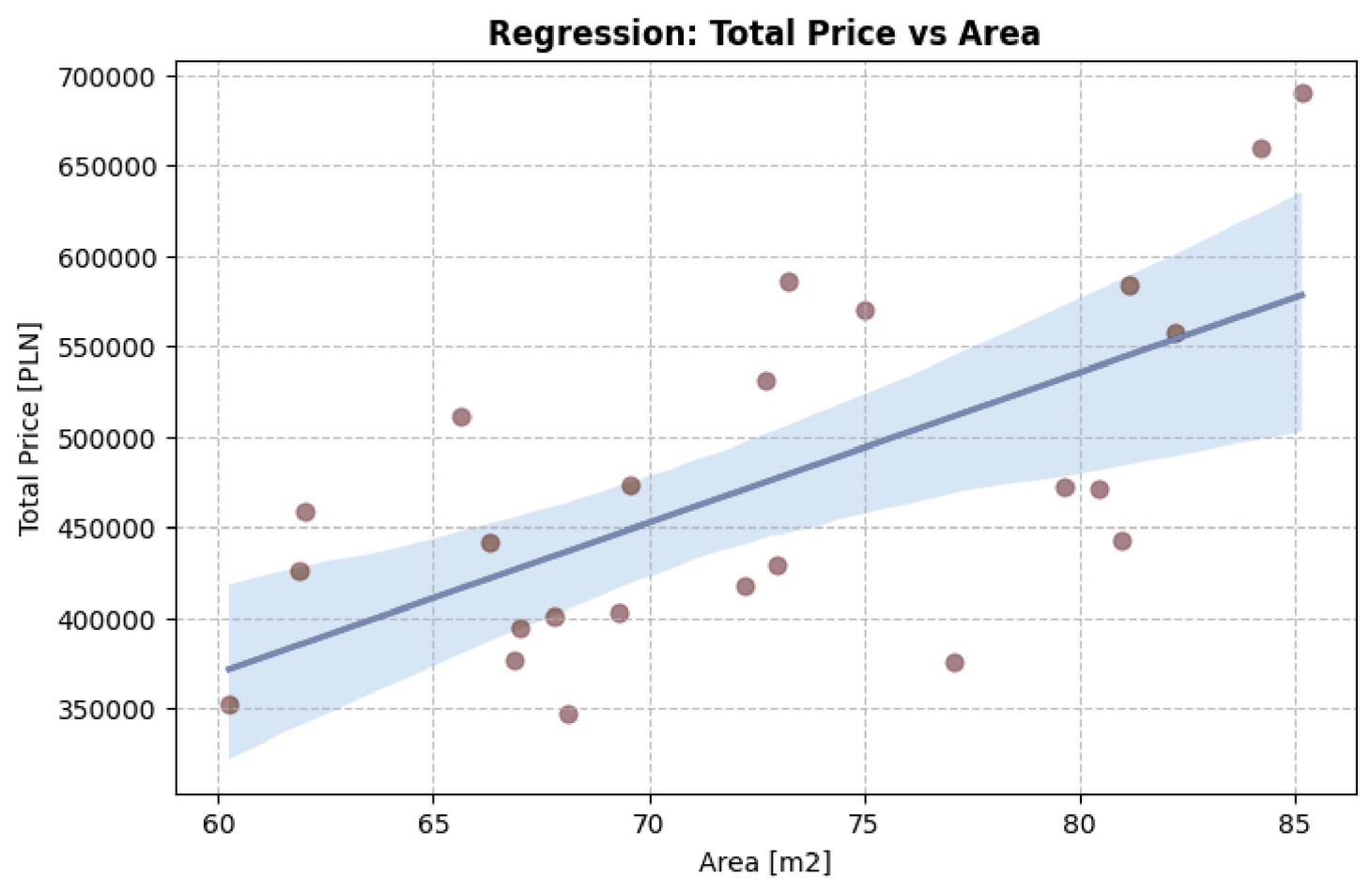
```
TREND: Daily price increase: 2.40 PLN/m2
Estimated annual increase: 877.81 PLN/m2
```

```
plt.figure(figsize=(8,5))
sns.regplot(
    x='Date_num',
    y='Unit Price [PLN/m2]',
    data=df,
    line_kws={'color': 'red'}
)
plt.title('Price Growth Trend Over Time', fontweight='bold')
plt.xlabel('Days from the start of analysis')
plt.ylabel('Unit price [PLN/m2]')
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```

Property Price vs. Area

Examined the relationship between the property size and its total price.

- **Method:** Simple linear regression (Total Price ~ Usable Area).
- **Result:** The coefficient of determination (R^2) is 0.43.
- **Finding:** Each additional square meter increased the total price by an average of 8,300 PLN.
- **Conclusion:** While the area is a significant factor, the relatively low R^2 value (43%) proves that in Krakow, location dictates the price to a much greater extent than the square footage.

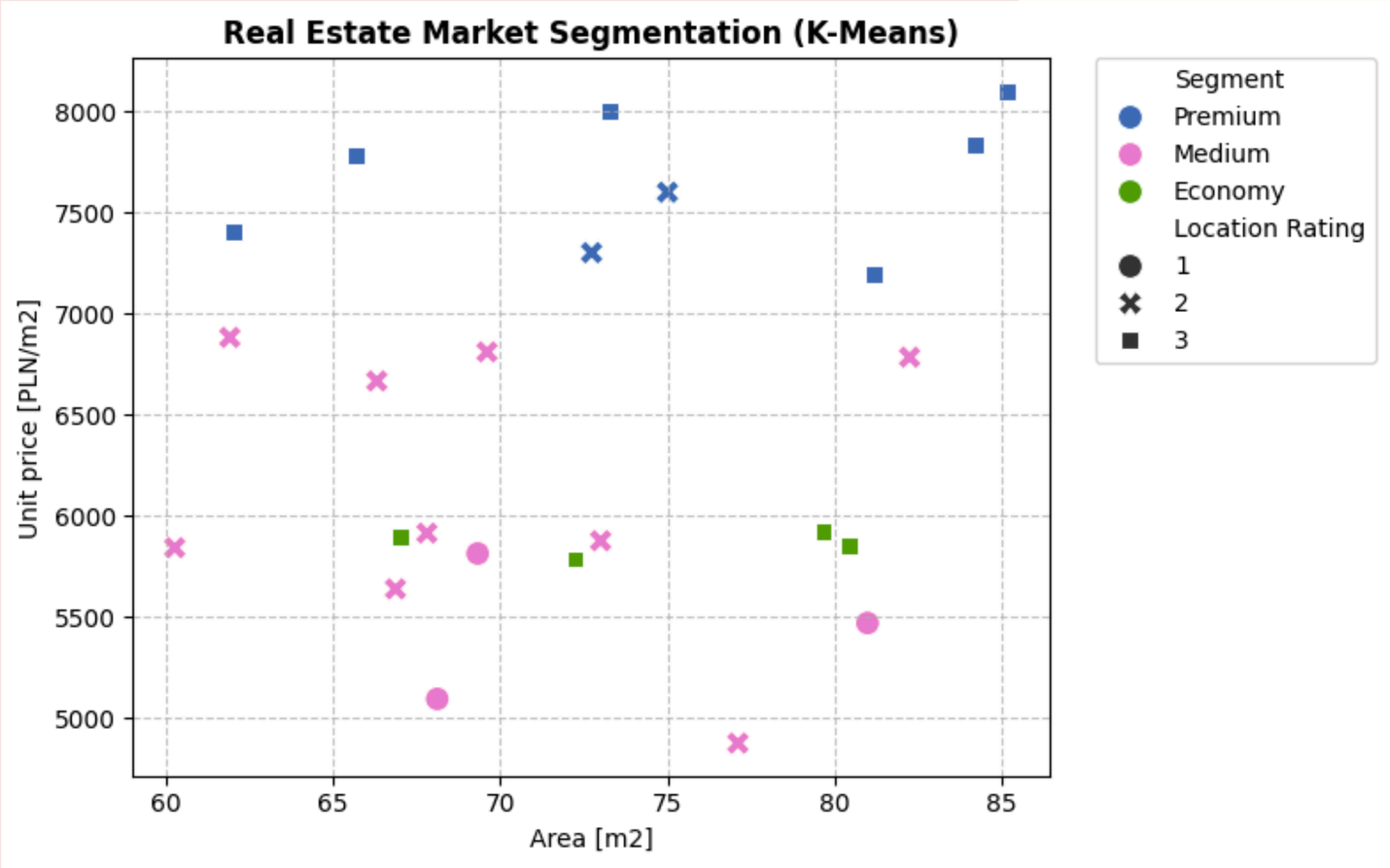


Implementation

```python
X_area = df[['Usable Area [m2]']]
y_total = df['Total Price [PLN]']

model_area = LinearRegression()
model_area.fit(X_area, y_total)
r2_area = model_area.score(X_area, y_total)

print(f"\nREGRESSION: Total Price vs Area")
print(f"Each m2 increases the property price by: {model_area.coef_[0]:.2f} PLN")
print(f"Model fit (R2): {r2_area:.2f}")
```

```
REGRESSION: Total Price vs Area
Each m2 increases the property price by: 8299.59 PLN
Model fit (R2): 0.43
```

```python
plt.figure(figsize=(8,5))
sns.regplot(
    x='Usable Area [m2]',
    y='Total Price [PLN]',
    data=df,
    scatter_kws={'s': 40},
    line_kws={'color': 'red'}
)
plt.title('Regression: Total Price vs Area', fontweight='bold')
plt.xlabel('Area [m2]')
plt.ylabel('Total Price [PLN]')
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



Property Classification (Market Segmentation)

Instead of subjective categorization, the K-Means Clustering machine learning algorithm was applied to divide the market into 3 natural groups.

- **Economy Segment (~5,800 PLN/sqm):** Properties in less favorable locations or requiring major renovation (e.g., Cechowa, Mogilska streets).
- **Medium Segment (~6,000 PLN/sqm):** Standard commercial spaces in residential neighborhoods.
- **Premium Segment (~7,700 PLN/sqm):** Properties in the strict city center (Długa, Starowiślna streets) and spaces leased to retail chains.

Implementation

```python
X_clus = df[['Unit Price [PLN/m2]', 'Location Rating']].copy()
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_clus)

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['Segment_ID'] = kmeans.fit_predict(X_scaled)

segment_means = df.groupby('Segment_ID')['Unit Price [PLN/m2]'].mean().sort_values()

name_map = {
    segment_means.index[0]: 'Economy',
    segment_means.index[1]: 'Medium',
    segment_means.index[2]: 'Premium'
}

df['Segment'] = df['Segment_ID'].map(name_map)

print("\nCLASSIFICATION (Average prices in groups):")
print(df.groupby('Segment')['Unit Price [PLN/m2]'].mean().sort_values())
```

```
CLASSIFICATION (Average prices in groups):
Segment
Economy    5865.362500
Medium     5972.473333
Premium    7652.443750
Name: Unit Price [PLN/m2], dtype: float64
```

```python
segment_order = ['Premium', 'Medium', 'Economy']
df['Segment'] = pd.Categorical(df['Segment'], categories=segment_order, ordered=True)

plt.figure(figsize=(8,5))
sns.scatterplot(
    x='Usable Area [m2]',
    y='Unit Price [PLN/m2]',
    hue='Segment',
    style='Location Rating',
    s=100,
    data=df,
    hue_order=segment_order
)

plt.title('Real Estate Market Segmentation (K-Means)', fontweight='bold')
plt.xlabel('Area [m2]')
plt.ylabel('Unit price [PLN/m2]')
plt.grid(True, linestyle='--', alpha=0.7)

plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', borderaxespad=0.)
plt.tight_layout()
plt.show()
```

Determining Market Feature Weights (Pricing Model)

A multiple regression model was built to evaluate the impact of individual attributes on the price per square meter.

$$Unit\ Price \approx 4712 + (642 \cdot Location) + (389 \cdot Standard) - (222 \cdot Accessibility)\ \ [PLN]$$

- **Method:** Multiple Linear Regression (Unit Price ~ Location + Standard + Accessibility).
- **Result:** The model determined the weights of specific attributes, indicating their exact impact on the price per sqm.
- **Conclusion:** Location proved to be the most significant factor increasing property value, with the finishing standard having an almost equally strong impact. The negative weight for transport accessibility stems from the specific nature of Krakow's city center – this is where the most expensive commercial spaces are located, despite limited vehicular access.

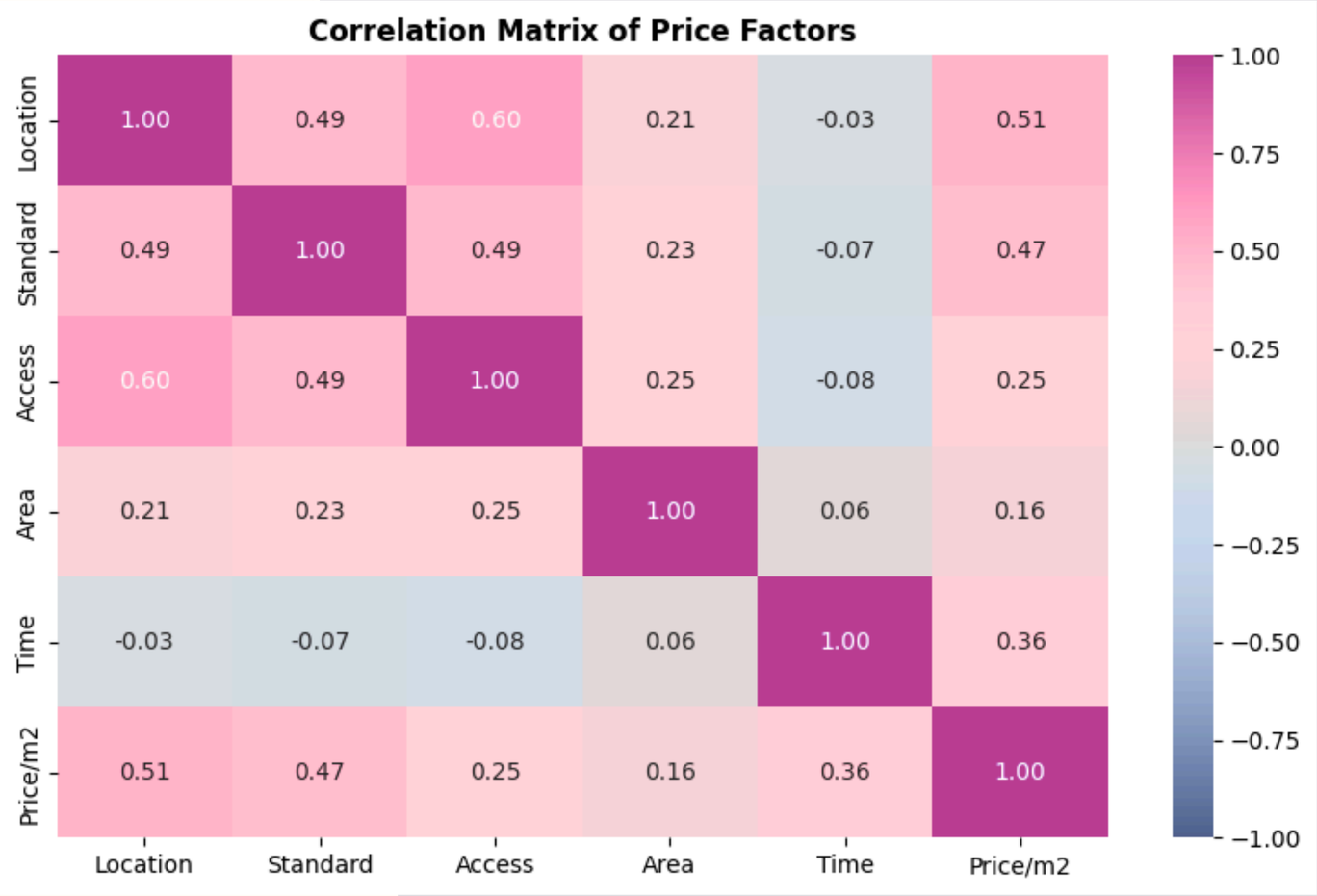
Implementation

```
X_multi = df[['Location Rating', 'Finishing Standard', 'Transport Accessibility']]
X_multi = sm.add_constant(X_multi)

y_multi = df['Unit Price [PLN/m2]']

model_weights = sm.OLS(y_multi, X_multi).fit()
print("\nFEATURE WEIGHTS (Multiple regression results):")
print(model_weights.params)
```

```
FEATURE WEIGHTS (Multiple regression results):
const                     4712.132690
Location Rating            642.540214
Finishing Standard         389.646622
Transport Accessibility   -222.356681
dtype: float64
```

Feature Correlation Analysis

- **Location Rating** (**0.51**): The strongest positive correlation. It confirms that the property's address remains the primary price determinant, although its dominance is not overwhelming.
- **Finishing Standard** (**0.47**): A very strong influence, closely following location. This indicates that the technical condition of the property was nearly as important to buyers as where it was situated.
- **Time** (**0.36**): A distinct positive correlation. This confirms the upward market trend, showing that prices rose systematically throughout the analyzed period.
- **Usable Area** (**0.16**): A weak correlation. The size of the property had a marginal effect on the price per square meter (the unit price did not fluctuate significantly with changes in square footage).



Implementation

```python
cols_corr = ['Location Rating', 'Finishing Standard',
             'Transport Accessibility', 'Usable Area [m2]',
             'Date_num', 'Unit Price [PLN/m2]']

cols_rename = {
    'Location Rating': 'Location',
    'Finishing Standard': 'Standard',
    'Transport Accessibility': 'Access',
    'Usable Area [m2]': 'Area',
    'Date_num': 'Time',
    'Unit Price [PLN/m2]': 'Price/m2'
}

corr_matrix = df[cols_corr].rename(columns=cols_rename).corr()

plt.figure(figsize=(10,6))
sns.heatmap(
    corr_matrix,
    annot=True,
    cmap='coolwarm',
    vmin=-1,
    vmax=1,
    center=0,
    fmt=".2f"
)
plt.title('Correlation Matrix of Price Factors', fontweight='bold')
plt.show()
```

```python
print("CORRELATION COEFFICIENTS WITH PRICE/M2")
print(corr_matrix['Price/m2'].sort_values(ascending=False))
```

```
CORRELATION COEFFICIENTS WITH PRICE/M2
Price/m2    1.000000
Location    0.509170
Standard    0.466390
Time        0.364093
Access      0.254474
Area        0.159862
Name: Price/m2, dtype: float64
```

Summary & Key Takeaways

1. **Prices grew steadily**, as confirmed by the positive temporal trend.
2. **Location was the primary valuation driver**, vastly outweighing the property size.
3. **The usable area had a limited impact on the price per square meter** - the exact address and its commercial potential are what truly dictate the value.
4. **Market segmentation** identified three distinct tiers: Economy, Medium, and Premium, highlighting a massive price advantage for the city center.
5. **The valuation model** confirmed that prices are primarily influenced by location, finishing standard, and the limited transport accessibility typical of the city center (negative correlation).
6. **Feature correlations** demonstrated that both the geographic placement and technical condition exert a very strong influence on the unit price.

Thank you for your attention!

Sources & References

Images:

- https://www.polska.travel/krakow/
- https://www.bip.krakow.pl/?dok_id=223498

Dataset:

- *Secondary market for commercial premises (retail and service units) in Krakow – analysis period: January 2017 to June 2018.*

Other Tools:

- Google Gemini